j-invariant, Elliptic Curves, and Moduli Spaces

Ryan Lam

The University of Arizona
(Based at the University of Bristol)

MATH 520B - Complex Analysis
May 2023

Introduction

This presentation introduces the j-invariant and two different ways of seeing it. We also introduce the "moduli space" of elliptic curves. The whole discussion is very algebra-geometric, and some of the main theorems are true in the more general setting (elliptic curves defined over a field K), here are some references I used:

- Silverman, J. H. (2014). *The Arithmetic of Elliptic Curves.* Springer.
- Prof. Richard Borcherds' YouTube video on Modular forms. `https://www.youtube.com/watch?v=f8pjt9ivjjc&list=PL8yHsr3EFj51HisRtNyzHX-Xyg6I3Wl2F`
- And of course, the recommended textbook: Donaldson, S. K. (2011). *Riemann Surfaces.* Oxford University Press.

Table of Contents

1. Theory of Riemann Surfaces
   - Important Theorems
2. The j-invariant: First Look
   - Definition Introduced in Class
   - $\mathcal{M}$, the "Moduli Space" of Elliptic Curves
3. The j-invariant: Second Look
   - Cross Ratios and the Lambda Modular function

What is an Elliptic Curve?

Definition

An **elliptic curve** is a *compact* Riemann surface of the quotient topological space $\mathbb{C}/L$, where L is a (non-degenerate) lattice $L := \mathrm{span}_{\mathbb{Z}}\langle \omega_1, \omega_2 \rangle, \omega_2/\omega_1 \notin \mathbb{R}$, with a marked point 0.

How to see the quotient space as a Riemann Surface?

Theorem (Donaldson, Section 3.2)

Given a Riemann surface X and a group Γ acts on X, if:

- *Around each point p of X we can find an open neighbourhood N such that if $q_1, q_2 \in N$ and $g \in \Gamma$ with $\gamma(q_1) = q_2$ then we must have $\gamma = 1$, $q_1 = q_2$.*

Then there is a Riemann Surface structure, denoted by X/Γ.

Classification of Elliptic Curves: First Look (1)

Theorem

Any elliptic curve $\mathbb{C}/L$ can be identified by (the closure of) an algebraic curve in $\mathbb{CP}^2$ by the following Riemann Surface Isomorphism:

$$z \mapsto [1 : \wp'(z) : \wp(z)], \qquad 0 \mapsto [0 : 1 : 0]$$

where $\wp$ is the ***Weierstrass $\wp$-function****:*

$$\wp(z) := \frac{1}{z^2} + \sum_{\lambda \in L \setminus \{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

Moreover, the algebraic follows the polynomial relation:

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

and the polynomial on the right hand side have three distinct roots.

Classification of Elliptic Curves: First Look (2)

We showed that in the relation $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$:

$$g_2 = 60 \sum_{\omega \in L\setminus\{0\}} \omega^{-4}, \qquad g_3 = 140 \sum_{\omega \in L\setminus\{0\}} \omega^{-6}$$

, and hence:

$$g_2 = \lambda^4 \widetilde{g_2}, \qquad g_3 = \lambda^6 \widetilde{g_3}.$$

Theorem

Two elliptic curves $\mathbb{C}/L, \mathbb{C}/\widetilde{L}$ are isomorphic if and only if $L = \lambda\widetilde{L}$ form some $\lambda \in \mathbb{C}^*$.

Taking $u = \lambda^2$ we immediately get the following:

Isomorphisms in terms of Weierstrass Equation

Theorem (Silverman, Prop. III.3.1(b))

If two elliptic curves $E, \widetilde{E}$ are realized as Weierstrass equation:

$$E : (\wp)'^2 = \wp^3 + A\wp + B$$

$$\widetilde{E} : \widetilde{(\wp)'}^2 = \widetilde{\wp}^3 + \widetilde{A}\widetilde{\wp} + \widetilde{B}$$

are related by a coordinate change $\widetilde{\wp} = u^2\wp, \widetilde{\wp'} = u^3\wp'$ for some $u \in \mathbb{C}^*$, then they are isomorphic (as Riemann Surface).

Table of Contents

1. Theory of Riemann Surfaces
   - Important Theorems
2. The j-invariant: First Look
   - Definition Introduced in Class
   - $\mathcal{M}$, the "Moduli Space" of Elliptic Curves
3. The j-invariant: Second Look
   - Cross Ratios and the Lambda Modular function

j-invariant: The First Definition

Recall the discriminant of the polynomial $x^3 + ax + b$ is $-4a^3 - 27b^2$, so in our case of $4x^3 - g_2x - g_3$, the discriminant is $\Delta = g_2^3 - 27g_3^2$.

Definition

The j-**invariant** of an elliptic curve $\mathbb{C}/L$ is:

$$j := 1728\frac{g_2^3}{\Delta} = 1728\frac{g_2^3}{g_2^3 - 27g_3^2}.$$

(Recall that g_2, g_3 depends on L.)

Changing $L \mapsto \lambda L$ would make j unchanged, so this is well defined up to the isomorphisms of elliptic curves.

An important property of j

Theorem (Silverman, Prop. III.1.4)

Let $E := \mathbb{C}/L, \widetilde{E} := \mathbb{C}/\widetilde{L}$ be two elliptic curves, then

1. *$j(E) = j(\widetilde{E})$ if and only if $E, \widetilde{E}$ are isomorphic as Riemann Surface, i.e. if and only if $L = \lambda\widetilde{L}$ for some $\lambda \in \mathbb{C}^*$.*
2. *For every $z \in \mathbb{C}$, there is E such that $j(E) = z$.*

Put it way easier...

Corollary

j is a bijection from $\mathcal{M} \to \mathbb{C}$, where $\mathcal{M}$ is the collection of all elliptic curves up to Riemann surfaces isomorphisms.

Proof Synopsis

Proof (Sketch).

1. We have shown the "$\Leftarrow$" direction just now. For "$\Rightarrow$", write

$$E : y^2 = x^3 + Ax + B,$$
$$\widetilde{E} : \widetilde{y}^2 = \widetilde{x}^3 + \widetilde{A}\widetilde{x} + \widetilde{B}.$$

   Then the assumption that $j(E) = j(\widetilde{E})$ yields $A^3\widetilde{B}^2 = \widetilde{A}^3 B^2$. We look for an isomorphism of the form $(x, y) = \left(u^2\widetilde{x}, u^3\widetilde{y}\right)$.

2. This has a quick proof:

$E : y^2 + xy = x^3 - \frac{36}{j_0 - 1728}x - \frac{1}{j_0 - 1728}$	$j = j_0$
$E : y^2 + y = x^3$	$j = 0$
$E : y^2 = x^3 + x$	$j = 1728$

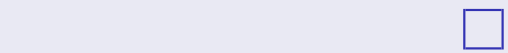


Table of Contents

1. Theory of Riemann Surfaces
   - Important Theorems
2. The j-invariant: First Look
   - Definition Introduced in Class
   - $\mathcal{M}$, the "Moduli Space" of Elliptic Curves
3. The j-invariant: Second Look
   - Cross Ratios and the Lambda Modular function

Visualizing The Moduli Space

Consider an elliptic curve C/L, $L := \mathrm{span}_{\mathbb{Z}}\langle \omega_1, \omega_2 \rangle$:

- If we look at the quantity $\tau = \omega_2/\omega_1$, this means:
  - $\tau \in \mathbb{H}$, and τ represent a lattice L up to scaling.
  - Also, this shows that for isomorphic elliptic curves, $\tau \in \mathbb{H}$ is unchanged.
- The $\mathcal{M}$ above can be "parametrized" by the quantity $\tau = \omega_2/\omega_1$ introduced above.
- However, this is not unique.

Question

Is there a "canonical choice" of τ?

It turns out there is, and this is the concept of a **fundamental domain**.

The "Canonical Choice"

Without loss of generality (since elliptic curves are isomorphic under scaling), we are asking the following question:

Question

Given lattices $L = \langle 1, \tau \rangle$ and $\widetilde{L} = \langle 1, \widetilde{\tau} \rangle$, when are they representing the same lattice?

There are two transformations:

$$\widetilde{\tau} = \tau - 1, \qquad \widetilde{\tau} = -\frac{1}{\tau}$$

It turns out any transformation, modulo scaling, is generated from those two.

A Picture of the Fundamental Domain

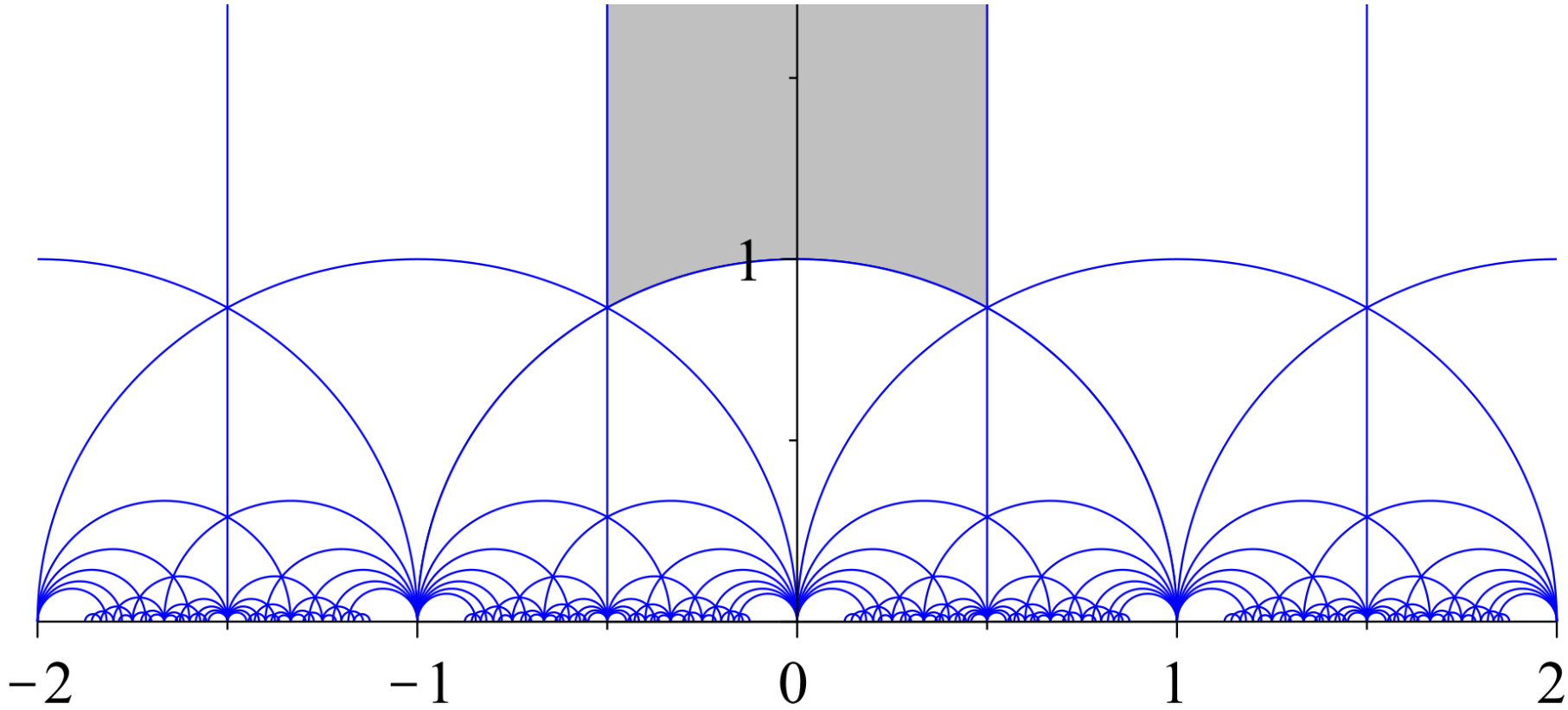


Figure: A Picture of the Fundamental Domain of j, credits here. An interactive animation can also be found at `https://roywilliams.github.io/play/js/sl2z/`.

Conclusion on $\mathcal{M}$

Theorem

The j-function is a bijective function from the fundamental domain Ω to $\mathbb{C}$ that satisfies:

$$j\left(\frac{a\tau+b}{c\tau+d}\right) = j(\tau) \qquad \text{for all } a,b,c,d \in \mathbb{Z}, ad-bc=1$$

Hence, this extends to a function $\mathbb{H} \to \mathbb{C}$ and is indeed a holomorphic function.

If we put on a Riemann Surface structure to the Fundamental Domain Ω, this will give us an isomorphism of Riemann surface, hence $\mathcal{M}$ will be isomorphic to $\mathbb{C}$ as Riemann surface.

Table of Contents

1. Theory of Riemann Surfaces
   - Important Theorems
2. The j-invariant: First Look
   - Definition Introduced in Class
   - $\mathcal{M}$, the "Moduli Space" of Elliptic Curves
3. The j-invariant: Second Look
   - Cross Ratios and the Lambda Modular function

$\wp$ as a Double Cover

Consider an elliptic curve $E = \mathbb{C}/\operatorname{span}_{\mathbb{Z}}\langle \omega_1, \omega_2 \rangle$:

Theorem (Riemann-Hurwitz)

If $f : X \to Y$ is a holomorphic map between Riemann Surface X, Y with genera g_X, g_Y respectively:

$$2 - 2g_X = (\deg f) \cdot (2 - 2g_Y) - \sum_{P \in X} (\operatorname{mult}_F(p) - 1)$$

Hence, consider the double cover $\wp : E \to \overline{\mathbb{C}}$, we have

$$\sum_{P \in E} (\operatorname{mult}_F(p) - 1) = 4.$$

So there are 4 distinct points with multiplicities of 2.

The Lambda Modular Function

In class, we showed the ramifications are actually

$$0, e_1 := \wp\left(\frac{\omega_1}{2}\right), e_2 := \wp\left(\frac{\omega_2}{2}\right), \text{ and } e_3 := \wp\left(\frac{\omega_1+\omega_2}{2}\right).$$

It is easy to check that e_1, e_2, e_3 are well defined with respect to $\tau = \omega_1/\omega_2$.

Definition

The quantity $\lambda \in \mathbb{C}$ with respect to an elliptic curve E (and the ratio $\tau \in \mathbb{H}$) is defined by:

$$\lambda(\tau) = \frac{e_3 - e_1}{e_2 - e_1}$$

Note that this is the cross ratio $\mathcal{X}(\infty, e_1, e_2, e_3)$ as 4 points in the Riemann sphere, and is indeed what we choose at the Riemann Existence theorem.

j-invariant: The Second Definition

Theorem (Silverman, Prop. III.1.7)

Given an elliptic curve $E = \mathbb{C}/L$:

$$j(E) = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

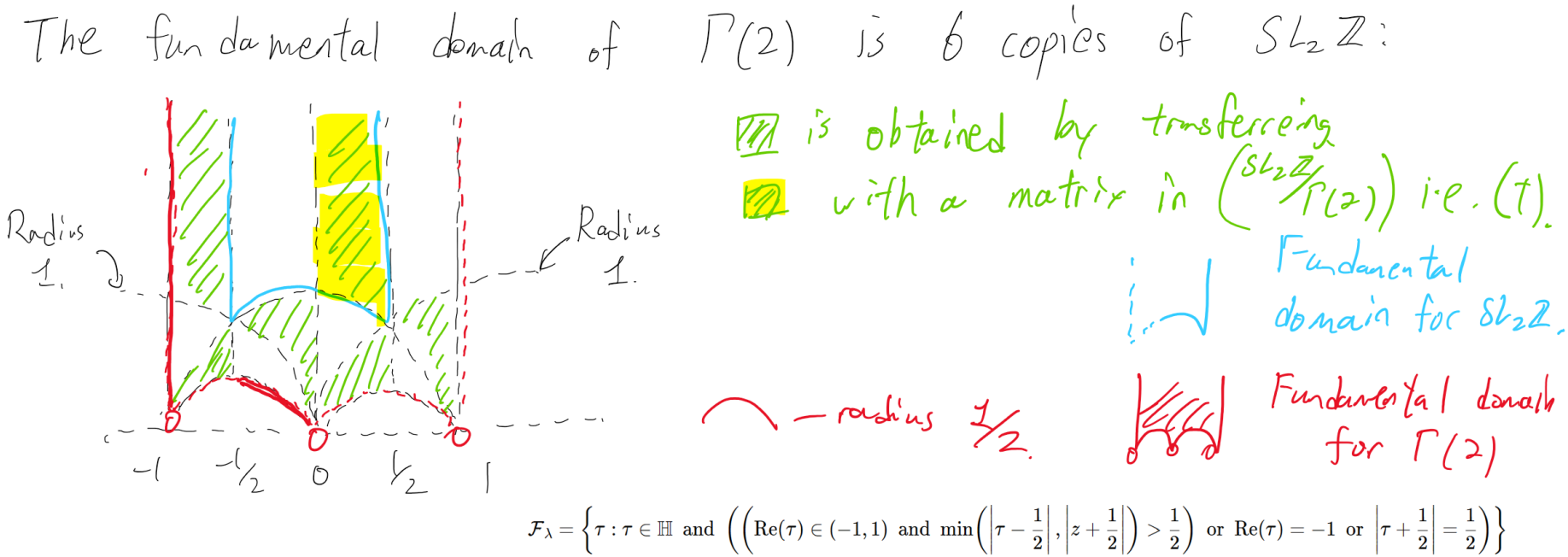


Figure: A Picture of the Fundamental Domain of λ.

Thank You!

Any questions?